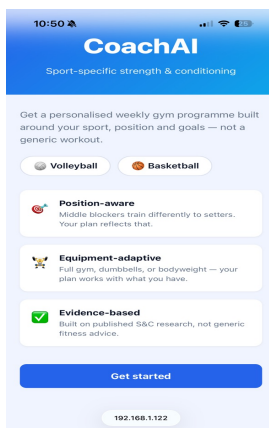


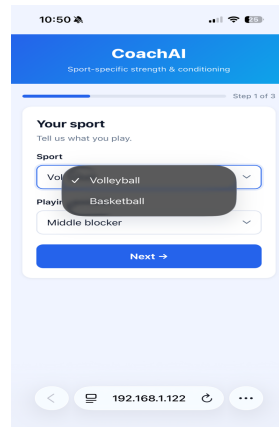
Abstract

CoachAI is a mobile-first web app that generates a personalised five-day strength and conditioning plan from sport, position, goal and equipment. It supports volleyball and basketball, uses a transparent rule-based recommender grounded in S&C literature, optional sign-in for saving plans, and an LLM-augmented beginner mode for novice users.

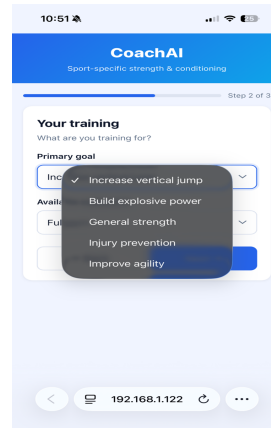
The System



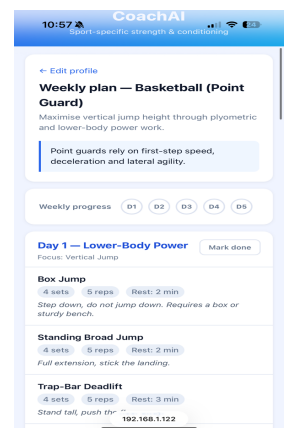
1. Landing page - explanation



2. Pick your sport and position (Step 2 of 3)



3. Pick a training goal (Step 2 of 3)



4. Generated plan with position note

Implementation

- **Backend.** Python + Flask. Single REST endpoint generates plans from a four-element user profile.
- **Recommender.** Transparent rule-based engine over a curated database of 40+ exercises tagged by sport, category and equipment level.
- **Session templates.** Five-day weekly structure — lower power, upper strength, lower strength, power & agility, core — with position notes and goal-driven adjustments.
- **Cycle 2: accounts.** Flask-Login + SQLite for optional sign-up, per-user plan saves, and persistent session-complete tracking.
- **Cycle 2: LLM mode.** Google Gemini rephrases coaching cues for novice users at development time; runtime serves a static cache — deterministic and offline-capable.

Conclusion & Future Work

An iterative two-cycle design — a rule-based prototype (Assignment 3) evaluated with amateur athletes, followed by an extended final system (Assignment 4) addressing the most-cited Phase 1 findings — produced a transparent, mobile-first S&C recommender that meets six of seven success criteria. Findings are reported honestly: criterion 4 (clarity $\geq 4/5$) was not met for novice users, motivating the new beginner-friendly explanations mode.

Introduction & Background

Amateur athletes (such as Goldsmiths' Volleyball team) lack qualified S&C support due to no funding. Existing AI fitness apps (FitnessAI, AI Endurance, Fitbit/Gemini) optimise general training or endurance sports — none address position-specific demands of indoor team sports and they also cost money or extra equipment. CoachAI fills this gap with a transparent, evidence-based recommender for sports where vertical jump and change-of-direction drive performance.

Testing & Evaluation

120/120

input combinations
produced valid plans

4.3 / 5

mean rating —
exercises feel
sport-relevant

N = 10

amateur athletes
in Phase 1 study

- **Functional.** Every supported (sport $\times$ position $\times$ goal $\times$ equipment) combination produces a structurally valid plan. Met criteria 1–3.
- **Sport relevance.** 90% of survey participants rated the plan's exercises as relevant to their sport ($\geq 4/5$). Met criterion 5 in absolute terms.
- **Clarity.** Mean 3.30/5; weaker for beginners (2.25) than advanced (4.50). Drove the Cycle 2 beginner-friendly LLM mode.
- **Demand for persistence.** 3 of 10 testers asked to save plans / track progress — directly motivated the Cycle 2 account layer.

- **Future work:** exercise demonstrations / videos for novices.
- Wider basketball-specific coverage.
- Richer goal-driven differentiation between plans.
- Longitudinal study over multiple weeks of training.